CoSteer: Federated Steering Beyond Local Attribute Coverage

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Abstract

Fine-grained control of language models is challenging when a client's training examples cover only a limited range of attribute levels. On condition that other clients may provide complementary supervision, we introduce Coverage-Aware Federated Steering (CoSteer), a personalized federated approach that jointly learns a shared steering direction and private client adapters. To map requested attribute levels into intervention strengths in the weight space, CoSteer learns a global nonlinear calibration curve through coverage-aware pooling, grounded by local attribute evidence. This design uses supervision across the federation to support control beyond individual clients' local attribute coverage while keeping their local adaptations effective. We demonstrate CoSteer on a news summarization task, examining extractiveness control, summary quality, and preservation of publication-specific characteristics. We also show that the learned control mechanism benefits new clients beyond the training federation.

1 Introduction

The usefulness of a language model depends on both the quality of its responses and how well they match a user's needs. Preferences such as the level of detail or formality often vary in degree, motivating *controlled generation*: specifying an output attribute and its desired level while maintaining task quality. Steering offers one approach by applying an intervention along a learned behavioral direction with adjustable strength (Rimsky et al., 2024; Cao et al., 2024; Fierro and Roger, 2026). Once learned, this direction allows the model's behavior to be adjusted without retraining for each requested attribute. Useful control requires these adjustments to produce fine-grained and predictable changes in the generated text.

Learning such control is difficult when training examples cover only a limited part of the desired attribute range. Different clients, however, may hold complementary examples. In a news article summarization task, for instance, summaries from different publications vary in how much they reuse words and phrases from the source article (Grusky et al., 2018). A publication whose summaries are mostly extractive may have little supervision for more abstractive outputs, even when such examples are available elsewhere. This motivates learning across clients, to extend control beyond each client's local coverage using the attribute range represented across the federation, while keeping training examples local.

Collaboration must also accommodate local specialization. Clients may share an attribute of interest while differing in their writing conventions, and a useful steering mechanism should work within these locally adapted models. Personalized federated learning combines shared knowledge with client-specific adaptation (Collins et al., 2021; Guo et al., 2025). For controlled generation, this decomposition must additionally support fine-grained variation across attribute levels, including those poorly represented in a client's own data. The challenges extend beyond learning a shared direction, as the model's response to its strength may be nonlinear, and each client provides supervision over a different part of the attribute range. These considerations lead to our central question: *Can clients with complementary attribute coverage jointly learn a steering mechanism that supports fine-grained control in their locally adapted models?*

We introduce CoSteer, a personalized federated approach that jointly learns a shared steering direction and private client adapters. Each client trains on its own reference outputs with steering active at their observed attribute levels. The direction is therefore learned in the presence of the private adaptations with which it will be used. To map requested levels to intervention strengths, CoSteer also learns a nonlinear calibration curve. Clients

contribute local calibration functions, whose values are pooled according to the evidence available near each attribute level. This coverage-aware pooling lets complementary clients inform different parts of the curve. An isotonic projection within the forward pass makes the resulting calibration monotone. Together, these components allow task adaptation and attribute control to be learned through the same training objective. Our contributions are:

1. CoSteer presents a protocol for learning a shared steering direction together with private client adapters. It is designed to extend control beyond individual clients' observed attribute ranges while retaining local adaptation.

2. We introduce coverage-aware calibration that pools local function according to attribute-level evidence and applies an isotonic projection during training. The resulting curve maps requested attribute levels to ordered intervention strengths.

3. We evaluate CoSteer on a news summarization task, examining control of summary extractiveness, generation quality and faithfulness, and preservation of publication-specific characteristics. We also study reuse of the learned control mechanism by clients beyond the training federation.

2 Methodology

2.1 Personalized Weight Steering

Problem setup. We consider a federation of I clients. Client $i \in \{1, \dots, I\}$ holds a dataset $\mathcal{D}^{(i)} = \{(x_n^{(i)}, y_n^{(i)}, \alpha_n^{(i)})\}_{n=1}^{N^{(i)}}$, where $x_n^{(i)}$ is an input, $y_n^{(i)}$ is a reference output, and $\alpha_n^{(i)} \in [0, 1]$ is $y_n^{(i)}$'s attribute level. Attribute levels are under a shared normalization, so the same value has a common interpretation across clients. Local datasets may cover different parts of this range. Our goal is to use their complementary supervision to support control at levels that are poorly represented or absent locally but represented elsewhere in the federation, while retaining client-specific adaptation.

Model parameterization. For a linear layer with frozen backbone weight W_0, client i uses $W_i(\alpha) = W_0 + W_i^P + \bar{g}(\alpha)\, W^D$ where W_i^P is a private adapter, W^D is a shared steering direction, and $\bar{g} : [0, 1] \to [0, 1]$ is a learnable calibration curve defined in Sec. 2.3.The private adapter captures local task characteristics, while the shared direction provides an attribute-dependent adjustment. Calibration determines the strength of this adjustment in the weight space at the requested level.

Training objective. Let $p^{(i)}(y \mid x, \alpha)$ denote client i's conditional distribution over output sequences. Each client minimizes $\mathcal{L}_i = \mathbb{E}_{(x,y,\alpha)\sim\mathcal{D}^{(i)}}\left[-\log p^{(i)}(y \mid x, \alpha)\right]$ with respect to the private adapter, local copy of the steering direction, and local calibration parameters. Steering is active at each reference output's observed attribute level. Task adaptation and control are thus learned jointly, allowing the direction to be optimized together with the private adapters.

2.2 Local Calibration Functions

A requested attribute level need not correspond to a proportional change in intervention strength. We therefore learn a nonlinear calibration rather than fixing $\bar{g}$ to identity. Each client maintains a local function $g^{(i)}(\alpha) = (1 - w^{(i)})\alpha + w^{(i)}\left[1 - (1 - \alpha^{p^{(i)}})^{q^{(i)}}\right]$, with parameters $\psi^{(i)} := (w^{(i)}, p^{(i)}, q^{(i)})$, where $w_i \in [0, 1]$ and $p^{(i)}, q^{(i)} > 0$. This function interpolates between the identity and the cumulative distribution function of a Kumaraswamy distribution (Kumaraswamy, 1980). It is monotone, satisfies $g^{(i)}(0) = 0$ and $g^{(i)}(1) = 1$, and accommodates different curve shapes with only three parameters. B.1 elaborates the function class.

2.3 Coverage-Aware Calibration

The parameters of $g^{(i)}$ influence the shape across the *entire* domain. Averaging these parameters across clients to obtain the global calibration curve $\bar{g}(\alpha)$ would allow each client to influence the curve even at attribute levels for which it has no supervision. Instead, we retain local calibration parameters and pool the function values using evidence near client's local attribute levels.

Local evidence. We evaluate the local functions on a common grid $\alpha_k = (k - 1)/(K - 1)$ for $k \in \{1, \dots, K\}$, with $K \geq 2$. Client i's evidence weight at grid point α_k is $c_k^{(i)} = \sum_{n=1}^{N^{(i)}} \max\left(0, 1 - \frac{|\alpha_n^{(i)} - \alpha_k|}{b}\right)$, where $b > 0$ is a fixed bandwidth. This triangular-kernel count assigns more weight to examples closer to α_k and zero weight to those at distance b or greater. The evidence weights are static throughout training.

Pooling function values. Let $\Omega_k = \sum_{i=1}^{I} c_k^{(i)}$ be the total evidence at α_k, and let $\mathcal{C} = \{k : \Omega_k > 0\}$ denote the covered grid indices. For $k \in \mathcal{C}$, we define the pooled function value $m_k = \frac{1}{\Omega_k} \sum_{i=1}^{I} c_k^{(i)} g_i(\alpha_k)$. Thus, each client's direct con-

tribution to the pooled function value is determined by its local evidence at that grid point, and a client with $c_k^{(i)} = 0$ does not contribute to m_k.

Monotone projection. Although every local function $g^{(i)}$ is monotone, the pooling weights vary across grid points, so the pooled sequence $\{m_k\}$ need not be monotone. We restore the monotonicity through weighted isotonic regression. Writing the covered indices as $k_1 < \cdots < k_M$, we solve

$$z^\star = \underset{(z_k)_{k\in\mathcal{C}}}{\arg\min} \sum_{k\in\mathcal{C}} \Omega_k (z_k - m_k)^2, \quad \text{subject to } z_{k_1} \le \cdots \le z_{k_M}. \tag{1}$$

The weights Ω_k give greater influence to pooled values supported by more evidence when resolving violations of monotonicity.

We set $\bar{g}(\alpha_k) = z_k^\star$ on covered grid points and fix the endpoints to $\bar{g}(0) = 0$ and $\bar{g}(1) = 1$. These endpoint values are consistent with the local functions and, when covered, with the projected sequence. Uncovered interior grid points are filled by linear interpolation between the nearest defined points on either side, including the fixed endpoints. Between grid points, $\bar{g}$ is also linearly interpolated. If no interior grid point is covered, $\bar{g}$ falls back to the identity function.

2.4 CoSteer (Alg. 1) Training Protocol

Let W_i^D and $g^{(i)}(\alpha_k)$ denote client i's steering direction and calibration values at the end of local training, and let $\Omega_k^{(-i)} = \sum_{j\neq i} c_k^{(j)}$ be the total evidence from the other clients. At the start of the next communication round, client i receives the aggregated steering direction $W^D \leftarrow I^{-1}\sum_{i=1}^{I} W_i^D$ and a held-out calibration average that excludes its own contribution. The server computes the held-out average for client i, $\bar{g}_k^{(-i)} = \frac{\sum_{j\neq i} c_k^{(j)} g^{(j)}(\alpha_k)}{\Omega_k^{(-i)}}$ if $\Omega_k^{(-i)} > 0$, and sets $\bar{g}_k^{(-i)} = 0$ otherwise. For calibration, the server sends client i a single table, each entry of which is an evidence-weighted average of the other clients' calibration function values at a grid point. During each local step, the client combines the received averages with its current function values to reconstruct $m_k^{(i)} = \frac{c_k^{(i)} g^{(i)}(\alpha_k) + \Omega_k^{(-i)} \bar{g}_k^{(-i)}}{\Omega_k}$, where $\Omega_k = c_k^{(i)} + \Omega_k^{(-i)}$. Monotone projection is then applied to obtain the current calibration curve $\bar{g}^{(i)}$ for the local forward pass. The local gradients pass through the isotonic regression to the current client's parameters ψ_i (Appendix B.2), along with the private adapter and the local steering direction.

3 Case Study on Extractiveness Control

Problem Setup. We study news summarization on Newsroom dataset (Grusky et al., 2018), and steer the *extractiveness* of the generated summary. Extractiveness is measured by fragment density (Grusky et al., 2018). Let $\mathcal{F}(A, S)$ be the set of maximal shared token sequences (extractive fragments) between article A and summary S. Then

$$\text{density}(A, S) = \frac{1}{|S|} \sum_{f\in\mathcal{F}(A,S)} |f|^2. \tag{2}$$

Density is low when a summary rewrites its source, and high when it copies long spans. We reimplemented the scorer with a regex tokenizer. Its Spearman correlation with the density values distributed with Newsroom is 0.99. Articles are truncated to 400 words, and density is recomputed on the truncated article. Pairs whose summary depends on the removed text are dropped. The same scorer labels the training data and evaluates generated summaries.

Clients. Each publication is a client. We take the 12 publications with at least 5,000 usable pairs whose median densities are evenly spaced, from telegraph.co.uk (median 1.3) to nypost.com (32.0). In our training protocol. Eight *participants* train the shared control mechanism, and four *held-out* clients join afterwards.

Baseline.

3.1 RQ1: Does collaboration improve control beyond local attribute coverage?

We first measure whether supervision from complementary clients in the training federation improves control at requested levels that are poorly represented or absent in a client's local training data.

Metrics. Errors are the absolute difference between the requested and the achieved percentile of extractiveness, averaged over $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$ and over publications. *Worst*: the worst publication. *In-/Out-of-supp.*: requested levels inside / outside the range of the publication's own training data. *Reach*: the fraction of out-of-support requests whose output leaves the publication's own range in the requested direction. *Spearman*: the rank correlation between

Table 1: Steering metrics on Newsroom test set, averaged over 8 publications. CoSteer achieves the lowest percentile errors both in and out of support.

Method	Pct err ↓	Worst-client ↓	In-supp. ↓	Out-of-supp. ↓	Spearman ↑
CoSteer	**0.143**±0.001	**0.190**±0.002	**0.133**±0.001	**0.147**±0.002	0.933±0.002
CoSteer (w. identity g)	0.148±0.000	0.193±0.002	**0.133**±0.000	0.157±0.001	0.932±0.001
CoSteer (w. global avg g)	0.150±0.000	0.204±0.001	0.135±0.001	0.157±0.001	0.928±0.001
Fed-Share	0.171±0.001	0.192±0.001	0.183±0.002	0.154±0.001	**0.948**±0.001
Local training	0.178±0.001	0.266±0.005	0.153±0.000	0.186±0.002	0.885±0.002
Task Arithmetic	0.182±0.000	0.255±0.005	0.152±0.001	0.198±0.001	0.882±0.002
Activation steering	0.253±0.000	0.303±0.001	0.212±0.001	0.287±0.000	0.781±0.001
Prompting	0.340±0.001	0.353±0.003	0.242±0.000	0.467±0.001	0.073±0.003

Table 2: House style evaluation on test set. Mean ± std over 3 random seeds.

Method	Acc. ↑	P(own) ↑	Gap ↓
CoSteer	0.569±.002	**0.381**±.002	**0.106**±.001
FedAvg (shared ad.)	0.407±.002	0.293±.004	0.187±.002
Local training	**0.572**±.004	0.379±.003	0.125±.001
Real summaries	0.588±.001	0.387±.001	–

Table 3: averaged on 4 held-out publications. Percentile error (↓) with n training pairs per new publication.

Method	n=256	n=1024	all
CoSteer	**0.148**±.001	**0.135**±.001	**0.129**±.001
Fed-Share	0.165±.001	0.165±.001	0.165±.001
Local	0.275±.001	0.231±.001	0.166±.001

the requested level and the output's extractiveness, per article.

Table 1 shows that our method achieves the lowest average percentile error, with improvements across both in-support and out-of-support evaluations. It also attains the lowest reported worst-case error across all test samples. These results suggest that our method improves the accuracy of the requested control across heterogeneous publications, rather than obtaining a lower overall error at the expense of difficult cases. The improvement outside the observed support is particularly encouraging, indicating that effective steering extends beyond the attribute ranges represented in local training. Among all baselines, training a fully-shared model offers a slightly higher spearman correlation than our method, which also indicates the steerability achieved by jointly training among clients. However, FedAvg loses significantly on the control **accuracy** represented by the percentile error.

3.2 RQ2: Do private adapters preserve publication-specific characteristics?

We test the role of personalization separately from overall control accuracy and summary quality.

Metrics. *Acc.*: share of generated summaries a publication classifier assigns to the right publication. *P(own)*: its mean probability of the right publication. *Gap*: mean absolute standardized difference of 11 style features from the publication's own summaries. Real: the publications' own test summaries (ceiling).

Table 2 shows that CoSteer retains publication-specific characteristics at locally supported attribute levels. CoSteer publication-classification accuracy is close to that of local training, and the mean probability assigned to the correct publication is also comparable. Both measurements remain close to those of the publications' own summaries. In contrast, the FedAvg model is less identifiable by publication on both measures. Since the classifier is density-matched, this comparison helps account for differences in extractiveness when assessing publication identity. The style-feature comparison supports the same interpretation. CoSteer has the smallest discrepancy from the publications' reference summaries across the eleven measured features, improving on both local training and the shared-adapter baseline. Combined with its stronger control performance, these findings suggest that sharing a steering mechanism can coexist with local specialization.

3.3 RQ4: How do learned calibration and coverage-aware pooling contribute?

We explain the contribution of calibration after the full-method comparisons. Separate the benefit of a flexible mapping from the benefit of combining clients' evidence according to coverage.

The first three rows of Table 1 show that

coverage-aware calibration improves control over both identity calibration and the global-averaging calibration mechanism. Relative to both ablations, the reduction in error is most significant on requests outside local support.

The global-averaging calibration mechanism does not outperform the identity map, suggesting that a learnable nonlinear mapping alone is insufficient to improve control. This highlights the importance of *how* calibration is learned from heterogeneous client data. Our results support coverage-aware pooling as an effective way to combine local calibration functions according to the attribute-level supervision available to each client.

3.4 RQ5: Can the learned control mechanism help new clients outside federation?

Finally, we evaluate whether the control mechanism learned by the federation remains useful for clients that did not participate in federated training. We compare three approaches: CoSteer, which trains a new client's private adapter while keeping the transferred steering direction and calibration function frozen; Fed-Share, which applies the global federated model without further adaptation; and Local, which learns from the new client's data without reusing the federated control mechanism. Table. 3 reports control error averaged over four held-out publications under different local data budgets.

CoSteer achieves the lowest control error across all evaluated data budgets, with its largest advantage over the global Fed-Share model and local training in the most data-limited setting. Notably, adapting CoSteer with only 256 training pairs yields lower error than local training with all available data. This suggests that the transferred mechanism provides useful control information that would otherwise need to be learned from local examples. As more data becomes available, local training improves and the gap narrows, but CoSteer retains an advantage even with the full dataset. Together, these results support the reuse of a shared control mechanism alongside client-specific adaptation: the learned direction and calibration remain useful beyond the private adapters encountered during federated training.

Acknowledgments

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A CoSteer Algorithm

B On the Calibration Function

B.1 Function Class of Calibration Function

Let $\kappa_{p,q}(\alpha) := 1 - (1 - \alpha^p)^q$. The weight w controls the interpolation: $w = 0$ gives $h(\alpha) = \alpha$, while $w = 1$ gives $h(\alpha) = \kappa_{p,q}(\alpha)$. If $p = q = 1$, $h(\alpha) = \alpha$ for every w.

When $w > 0$ and $q = 1$, $\kappa_{p,1}(\alpha) = \alpha^p$: for $p > 1$, h is convex and lies below the identity function, assigning relatively less strength to lower requested levels. For $0 < p < 1$, it is concave and lies above the identity.

When $w > 0$ and $p = 1$, $\kappa_{1,q}(\alpha) = 1-(1-\alpha)^q$: for $q > 1$, h is concave and lies above the identity, while for $0 < q < 1$, it is convex and lies below the identity, deferring more of the change to higher requested levels.

When $w > 0$ and $p, q > 1$, h changes curvature from convex to concave; when $w > 0$ and $0 < p, q < 1$, the curvature changes in the reverse direction.

B.2 Backpropagation of Monotone Calibration Curve

(1) Interpolation. A training pair with level $\alpha \in [\alpha_k, \alpha_{k+1}]$ uses $\bar{g}_l^{(i)}(\alpha) = (1 - \rho)\, \bar{g}_l^{(i)}(\alpha_k) + \rho\, \bar{g}_l^{(i)}(\alpha_{k+1})$, where $\rho = (\alpha - \alpha_k)(K - 1)$. The coefficient's gradient, $\partial \ell / \partial \bar{g}_l^{(i)}(\alpha) = \sum_t \langle \partial \ell / \partial y_t,\, B^D A^D x_t \rangle$ summed over token positions t, is split between the two neighbouring grid values with weights $1 - \rho$ and ρ. Summed over the batch, this gives $G_{l,k} = \partial \mathcal{L}_i / \partial \bar{g}_l^{(i)}(\alpha_k)$. At a grid point that no client covers, Π sets the value to $(1 - t)\, \bar{g}_l^{(i)}(\alpha_{k^-}) + t\, \bar{g}_l^{(i)}(\alpha_{k^+})$, with $t = (\alpha_k - \alpha_{k^-})/(\alpha_{k^+} - \alpha_{k^-})$. This is part of the definition of Π, not an approximation, and the inference table uses the same rule. The value is a fixed linear function of its two neighbours, so it passes its gradient to them with weights $1 - t$ and t. Any value between the two neighbours keeps $\bar{g}$ monotone; linear interpolation additionally makes $\bar{g}$ a single straight line between α_{k^-} and α_{k^+}, as if the uncovered points were absent from the grid. On Newsroom no grid point is uncovered: every α_k has a total smoothed count above 3,000. The endpoints are the constants 0 and 1 and pass none.

(2) Projection. The solution of isotonic regression has a block structure, which defines $B(k)$. A *block* is a set of covered points that are consecutive in $\mathcal{C}$, $B = \{k_r, k_{r+1}, \ldots, k_s\}$; it may span uncovered grid points, which carry no weight. A block's weight and weighted mean are

$$\Omega_B = \sum_{k \in B} \Omega_k, \qquad \mu(B) = \frac{1}{\Omega_B} \sum_{k \in B} \Omega_k\, m_{l,k}^{(i)}. \tag{3}$$

The pool-adjacent-violators algorithm partitions $\mathcal{C}$ into blocks $B_1, \ldots, B_r$, ordered from left to right. It starts with every covered point as its own block. Scanning from left to right, whenever a block's mean is strictly greater than the mean of the block to its right, it merges the two, and it repeats the check with the merged block's left neighbour. It stops when $\mu(B_1) \le \mu(B_2) \le \cdots \le \mu(B_r)$. $B(k)$ denotes the block that contains grid point k, and the solution assigns every point of a block the block's mean:

$$\bar{g}_l^{(i)}(\alpha_k) = \mu\big(B(k)\big) = \frac{1}{\Omega_{B(k)}} \sum_{k' \in B(k)} \Omega_{k'}\, m_{l,k'}^{(i)}. \tag{4}$$

The partition depends on the current pooled values, so it is recomputed for every layer and every forward pass. We drop the indices i and l from B for readability. When the endpoints are covered (as on Newsroom), each forms a single-point block. Every shape satisfies $h(0) = 0$ and $h(1) = 1$, so $m_{l,1} = 0$ and $m_{l,K} = 1$ are the smallest and largest of the pooled values, and a strict violation involving them cannot occur. We compute the partition from the current values without gradient, and then differentiate the block means with the partition held fixed:

$$\frac{\partial \bar{g}_l^{(i)}(\alpha_k)}{\partial m_{l,k'}^{(i)}} = \mathbf{1}\big[k' \in B(k)\big]\, \frac{\Omega_{k'}}{\Omega_{B(k)}}. \tag{5}$$

This is the exact Jacobian of weighted isotonic regression wherever the partition does not change under small perturbations of $m_l^{(i)}$. That holds everywhere except on a measure-zero set of pooled values, where two adjacent groups of points have exactly equal weighted means; there the map is not differentiable (cf. the unweighted case in Blondel et al., 2020). Two cases follow. If $m_l^{(i)}$ is already increasing, every block is a single point and the Jacobian is the identity. If a block has merged several points, the gradient arriving at any of them is shared among all members in proportion to their weights. This is the only place where the projection is non-local.

(3) Pooling. By the definition of $m_{l,k}^{(i)}$, $\partial m_{l,k}^{(i)}/\partial h_{i,l}(\alpha_k) = \omega_{ik}/\Omega_k$. Combined with Equation 5, $\Omega_{k'}$ cancels:

$$\frac{\partial \bar{g}_l^{(i)}(\alpha_k)}{\partial h_{i,l}(\alpha_{k'})} = \mathbf{1}\big[k' \in B(k)\big]\, \frac{\omega_{ik'}}{\Omega_{B(k)}}. \qquad (6)$$

Client i's value at $\alpha_{k'}$ therefore receives gradient only if $\omega_{ik'} > 0$, that is, only where the client has data. Its share is its evidence relative to the total evidence of the block. Where several clients overlap, each moves the pooled value in proportion to its weight.

(4) Shape. The gradient reaches $\psi_{i,l}$ through the closed-form derivative of the Kumaraswamy mixture $h_{i,l}(\alpha_{k'})$, computed by automatic differentiation. In total,

$$\nabla_{\psi_{i,l}} \mathcal{L}_i = \sum_k G_{l,k} \sum_{k' \in B(k)} \frac{\omega_{ik'}}{\Omega_{B(k)}} \nabla_{\psi_{i,l}} h_{i,l}(\alpha_{k'}), \qquad (7)$$

where k runs over the covered grid points, and $G_{l,k}$ includes the gradient passed on by interpolated neighbours. The shape is parametric, so a step on $\psi_{i,l}$ also changes $h_{i,l}$ where the client has no data. Those values carry zero weight in $\bar{g}$, so they affect neither training nor inference. The projection is applied separately to each layer, so the loss at layer l trains only that layer's shapes.

C Additional Results on Newsroom dataset

C.1 Metrics Definition

Spearman. For paired observations $(x_i, y_i)_{i=1}^n$, Spearman's rank correlation is

$$\rho_s = \frac{\sum_{i=1}^n (R_i - \bar{R})(S_i - \bar{S})}{\sqrt{\sum_{i=1}^n (R_i - \bar{R})^2}\sqrt{\sum_{i=1}^n (S_i - \bar{S})^2}}$$

where $R_i = \mathrm{rank}(x_i)$ and $S_i = \mathrm{rank}(y_i)$. Ties receive their average rank.

Here, x_i is the control level and y_i is the measured attribute value. Thus, $\rho_s \in [-1, 1]$ measures how consistently the attribute increases with the control level. Your evaluation computes this per input, then reports

$$\mathrm{Spearman} = \frac{1}{N} \sum_{j=1}^{N} \rho_s^{(j)},$$

assigning 0 to inputs whose attribute values are constant across control levels.

C.2 Client-wise Analysis

The gains are largest where a client's own data does not cover the request. Table 4 breaks the comparison down by client. Collaboration therefore helps most where a client's data has gaps in coverage, consistent with the premise that clients with complementary coverage benefit from a shared direction.

C.3 RQ3: Is improved control accompanied by good summary quality and faithfulness?

We measure whether gains in extractiveness control preserve the usefulness and factual reliability of the summaries. We compare CoSTEER with locally trained model and real summaries, which are the publications' own test summaries of the same articles.

Metric. AlignScore: consistency with the article. BERTScore F1: similarity to the reference summary. LLM judge (Qwen2.5-7B-Instruct, every test summary): faithfulness is the fraction of summary sentences supported by the article; relevance and coherence are rated 1–5.

Table 5 shows that CoSTEER remains close to local training, and in reference similarity, relevance, and coherence when evaluated across requested levels. The comparison with real summaries also provides a reference ceiling that CoSTEER can match up with. These results suggest that the improvement in extractiveness control is accompanied by broadly relevant content selection and readability. Local training nevertheless retains an advantage on the reported quality measure, with the clearest difference appearing in AlignScore. Overall, the significant gains in control come with only a mild quality trade-off.

D Beyond Language: Steering Lipophilicity on Molecule Generation

Table 4: Per-publication steering on Newsroom (test set). Below each publication, its support: the 5th–95th percentile of its own training data on the global extractiveness scale. *In-/Out-of-supp.*: requested levels inside / outside that range. Publications are ordered by the lower end of their support. Best value per publication in bold.

Publication	Method	Pct err ↓	In-supp. ↓	Out-of-supp. ↓	Spearman ↑
forbes.com [0.03, 0.99]	Ours	0.129±0.001	0.151±0.002	**0.096**±0.000	0.934±0.003
	FedAvg (shared adapter)	0.160±0.002	0.177±0.004	0.133±0.003	**0.964**±0.001
	Local training	**0.127**±0.002	**0.142**±0.003	0.103±0.003	0.936±0.003
people.com [0.03, 0.70]	Ours	**0.142**±0.002	**0.107**±0.002	**0.166**±0.001	**0.920**±0.006
	FedAvg (shared adapter)	0.160±0.001	0.145±0.002	0.170±0.002	0.908±0.002
	Local training	0.171±0.000	0.113±0.002	0.209±0.001	0.891±0.004
wsj.com [0.04, 0.89]	Ours	**0.131**±0.002	**0.121**±0.001	0.146±0.006	0.944±0.002
	FedAvg (shared adapter)	0.169±0.002	0.188±0.002	**0.140**±0.004	**0.963**±0.002
	Local training	0.152±0.004	0.149±0.004	0.155±0.005	0.946±0.007
theguardian.com [0.04, 0.55]	Ours	**0.125**±0.000	**0.106**±0.001	**0.137**±0.000	0.951±0.005
	FedAvg (shared adapter)	0.167±0.003	0.192±0.007	0.150±0.001	**0.954**±0.002
	Local training	0.180±0.003	0.161±0.004	0.193±0.003	0.864±0.004
aol.com [0.06, 0.82]	Ours	**0.124**±0.001	**0.118**±0.002	**0.133**±0.002	0.949±0.002
	FedAvg (shared adapter)	0.171±0.003	0.179±0.004	0.160±0.002	**0.951**±0.005
	Local training	0.136±0.001	0.128±0.002	0.147±0.002	0.946±0.003
reuters.com [0.10, 0.97]	Ours	**0.190**±0.002	**0.212**±0.001	0.158±0.008	0.897±0.006
	FedAvg (shared adapter)	0.191±0.002	0.222±0.004	**0.144**±0.001	**0.947**±0.006
	Local training	0.213±0.002	0.236±0.005	0.179±0.011	0.842±0.008
cbc.ca [0.11, 0.95]	Ours	**0.136**±0.002	**0.120**±0.003	**0.159**±0.002	**0.956**±0.001
	FedAvg (shared adapter)	0.191±0.001	0.206±0.002	0.169±0.002	0.944±0.003
	Local training	0.182±0.001	0.169±0.003	0.200±0.004	0.920±0.003
nypost.com [0.51, 0.95]	Ours	0.167±0.001	**0.126**±0.004	0.178±0.000	0.914±0.011
	FedAvg (shared adapter)	**0.159**±0.001	0.153±0.002	**0.161**±0.002	**0.953**±0.003
	Local training	0.266±0.005	0.128±0.003	0.301±0.006	0.732±0.019

Table 5: Summary quality on model generation and real summaries. Mean ± std over 3 random seeds.

Method	AlignScore	BERTScore F1	Faithfulness	Relevance	Coherence
Ours	0.757±0.004	0.894±0.000	0.804±0.002	3.602±0.007	4.185±0.003
Local training	0.783±0.005	0.896±0.000	0.811±0.002	3.619±0.005	4.203±0.003
Real summaries	0.762±0.004	–	0.844±0.003	3.670±0.005	4.189±0.007

Algorithm 1 Coverage-Aware Federated Steering (COSTEER)

Input: Client datasets $\{\mathcal{D}^{(i)}\}_{i=1}^{I}$; frozen backbone W_0; communication rounds T; local steps E; grid size K; bandwidth b.
Output: Shared steering direction W^D, global calibration curve $\bar{g}$, and private adapters $\{W_i^P\}_{i=1}^{I}$.

Initialize W^D, private adapters W_i^P, and local calibration parameters $\psi^{(i)} = (w^{(i)}, p^{(i)}, q^{(i)})$, with $w^{(i)} \in [0,1]$ and $p^{(i)}, q^{(i)} > 0$.
Define $g^{(i)}(\alpha) = (1-w^{(i)})\alpha + w^{(i)}[1-(1-\alpha^{p^{(i)}})^{q^{(i)}}]$ and $\alpha_k = (k-1)/(K-1)$.
On each client i:
Compute fixed evidence weights $c_k^{(i)} = \sum_{n=1}^{N^{(i)}} \max\Big(0, 1 - |\alpha_n^{(i)} - \alpha_k|/b\Big)$.
Send $\{c_k^{(i)}, g^{(i)}(\alpha_k)\}_{k=1}^{K}$ to the server.
On the server: set $\Omega_k = \sum_{i=1}^{I} c_k^{(i)}$ and $\mathcal{C} = \{k : \Omega_k > 0\}$.
for $t = 1, 2, \ldots, T$ **do**
  On the server:
  for all clients i **do**
    Compute $\Omega_k^{(-i)} = \Omega_k - c_k^{(i)}$ and the held-out calibration table

$$\bar{g}_k^{(-i)} = \begin{cases} \dfrac{\sum_{j \neq i} c_k^{(j)} g^{(j)}(\alpha_k)}{\Omega_k^{(-i)}}, & \Omega_k^{(-i)} > 0, \\ 0, & \Omega_k^{(-i)} = 0. \end{cases}$$

    Send W^D and $\{\bar{g}_k^{(-i)}, \Omega_k^{(-i)}\}_{k=1}^{K}$ to client i.
  end for
  for all clients i **in parallel do**
    Set $W_i^D \leftarrow W^D$; retain W_i^P and $\psi^{(i)}$ from the previous round.
    for $e = 1, 2, \ldots, E$ **do**
      Sample a minibatch $\mathcal{B}_i \subset \mathcal{D}^{(i)}$.
      Reconstruct the pooled values using the current local function and fixed held-out table:

$$m_k^{(i)} = \frac{c_k^{(i)} g^{(i)}(\alpha_k) + \Omega_k^{(-i)} \bar{g}_k^{(-i)}}{\Omega_k}, \qquad k \in \mathcal{C}.$$

      Compute $\bar{g}^{(i)} \leftarrow \text{CALIBRATE}(m^{(i)}, \Omega, \mathcal{C})$.
      Compute the minibatch loss with steering active at each reference's attribute level:

$$\mathcal{L}_i = -\frac{1}{|\mathcal{B}_i|} \sum_{(x,y,\alpha) \in \mathcal{B}_i} \log p^{(i)}(y \mid x, \alpha), \qquad W_i(\alpha) = W_0 + W_i^P + \bar{g}^{(i)}(\alpha) W_i^D.$$

      Take a local optimizer step on $(W_i^P, W_i^D, \psi^{(i)})$ using $\nabla \mathcal{L}_i$, differentiating through calibration and preserving the parameter constraints.
    end for
    Send W_i^D and $\{g^{(i)}(\alpha_k)\}_{k=1}^{K}$ to the server; keep W_i^P and $\psi^{(i)}$ local.
  end for
  On the server: aggregate $W^D \leftarrow I^{-1} \sum_{i=1}^{I} W_i^D$ and cache the received calibration tables.
end for
Compute $m_k = \Omega_k^{-1} \sum_{i=1}^{I} c_k^{(i)} g^{(i)}(\alpha_k)$ for $k \in \mathcal{C}$, and set $\bar{g} \leftarrow \text{CALIBRATE}(m, \Omega, \mathcal{C})$.
return W^D, $\bar{g}$, and $\{W_i^P\}_{i=1}^{I}$.

function CALIBRATE($m, \Omega, \mathcal{C}$)
  Solve weighted isotonic regression on the covered indices $\mathcal{C} = \{k_1 < \cdots < k_M\}$:

$$z^\star = \underset{(z_k)_{k \in \mathcal{C}}:\, z_{k_1} \leq \cdots \leq z_{k_M}}{\arg\min} \sum_{k \in \mathcal{C}} \Omega_k (z_k - m_k)^2.$$

  Set $h(\alpha_k) = z_k^\star$ for $k \in \mathcal{C}$, and fix $h(0) = 0, h(1) = 1$.
  Fill uncovered grid points and evaluate between grid points by linear interpolation; use $h(\alpha) = \alpha$ if no interior grid point is covered.
  return h.
end function

Table 6: Knob calibration on pruned ChEMBL (E1), 8 participating clients, test split.

Method	percentile error ↓				Spearman ↑
	mean	worst-client	in-sup.	out-of-sup.	
CoSteer	$\mathbf{0.162} \pm 0.001$	$\mathbf{0.231} \pm 0.001$	$\mathbf{0.155} \pm 0.002$	$\mathbf{0.163} \pm 0.002$	$\mathbf{0.915} \pm 0.002$
CoSteer (w. identity g)	0.191 ± 0.001	0.270 ± 0.005	0.163 ± 0.001	0.218 ± 0.002	0.860 ± 0.008
CoSteer (w. global avg g)	0.180 ± 0.001	0.240 ± 0.003	0.162 ± 0.002	0.195 ± 0.003	0.880 ± 0.003
Fed-Share	0.177 ± 0.001	0.243 ± 0.002	0.162 ± 0.001	0.186 ± 0.001	0.902 ± 0.004
Local training	0.192 ± 0.002	0.289 ± 0.006	0.159 ± 0.001	0.218 ± 0.004	0.838 ± 0.005
Task Arithmetic	0.249 ± 0.001	0.308 ± 0.000	0.197 ± 0.002	0.301 ± 0.002	0.723 ± 0.004
Activation Steering	0.364 ± 0.000	0.418 ± 0.001	0.227 ± 0.002	0.497 ± 0.000	-0.098 ± 0.003
Prompting	0.352 ± 0.001	0.417 ± 0.004	0.224 ± 0.001	0.477 ± 0.002	0.044 ± 0.005